WBC wrist-tracking stabilization: reference smoothing + stiffness-scaled damping (τ=0.08 s, slew=3.0 rad/s, D=0.6+0.15√K, K=30)
E1 noisy wrist track
Right Arm 4 [deg from start]
time [s]
commanded (jittery IK)
realized, smoother OFF (3 rad/s²)
realized, smoother ON (1 rad/s²)
intended sweep
E2 sudden wrist jump
step response [deg]
time [s]
commanded step
OFF (overshoot 31%)
ON (overshoot 11%)
E3 buzz reduction
(clean-sweep added lag 44 ms)
realized joint accel RMS [rad/s²]
smoother OFF
smoother ON
noisy track
step jump